

Figure 4.8 Northrop Grumman X-47B (2012).⏎

4.5.2.8 Wing Loong I (2015)

The Wing Loong I features a conventional MALE UAV configuration, with a slender fuselage, mid-mounted straight wings, and a rear-mounted pusher propeller. Constructed primarily from composite materials, its airframe is designed for durability, low weight, and aerodynamic efficiency. The UAV has a wingspan of 14 meters (46 feet), a length of 9 meters (29.5 feet), and can achieve endurance up to 20 hours with an operational ceiling of 16,400 feet (5000 meters).

Powered by a Rotax-type piston engine, the Wing Loong I has a maximum speed of 280 km/h (174 mph) and a range of 4000 km, enabling extended missions for surveillance, reconnaissance, and tactical strike operations. The aircraft is equipped with a retractable tricycle landing gear, allowing for autonomous take-off and landing on semi-prepared runways (Hwang, 2020).

Avionics and sensor systems

The Wing Loong I's modular payload bay supports a wide array of ISR and combat sensors, including: